Discrete-Time Low-Pass Filter Design

1.0 Learning Outcomes……1
2.0 Introduction……1
3.0 Discrete-Time Low-Pass Filter Design……...2
4.0 Summary……...7
Recommended Citation……7

1.0 Learning Outcomes

After studying this ACE-Lab learning guide, you should be able to demonstrate the Learning Outcomes listed in Table 1. On completion of the ACE-Lab learning guide, revisit Table 1 and confirm whether each Learning Outcome has been achieved.

Table 1: *Learning Outcomes from the Discrete-Time Low-Pass Filter Design Learning Guide*

ID	Learning Outcome	Achieved
1.	Explain the purpose and operation of a first-order discrete-time low-pass filter.	☐
2.	Select suitable filter coefficients to balance measurement smoothing and response speed while ensuring unity steady-state gain and stable filter behaviour.	☐
3.	Implement and evaluate a discrete-time low-pass filter, considering the effects of sampling interval and initialisation.	☐

2.0 Introduction

Measurements obtained from physical systems often contain unwanted high-frequency variation caused by sensor noise, electrical interference, quantisation, vibration or other measurement effects. If these rapid variations are passed directly to a controller, they can influence the calculated control action and may result in unnecessary or rapidly changing actuator commands. Measurement noise can also make experimental data more difficult to interpret when evaluating system behaviour.

As illustrated in Figure 1, a discrete-time low-pass filter can be applied to the measured signal, $x[k]$, to produce a filtered signal, $y[k]$, before the measurement is used for control or system evaluation. The purpose of the filter is to reduce unwanted high-frequency variation while retaining the underlying changes in the measurement that are important for understanding and controlling the physical system.

A low-pass filter can therefore be used to reduce these high-frequency variations while retaining the slower changes that represent the behaviour of interest. However, filtering introduces an important practical trade-off. Increasing the amount of smoothing can reduce measurement noise more effectively, but it also causes the filtered signal to respond more slowly to genuine changes in the measured quantity. The filter must therefore be selected so that noise is reduced without introducing excessive delay into the measurement used for control.

This learning guide introduces the design and implementation of a first-order discrete-time low-pass filter. The filter is developed directly in the discrete-time domain so that it can be implemented using a difference equation on a microcontroller or within a discrete-time control system. No mathematical model of the physical system is required. Instead, the filter is selected and refined by examining its measured behaviour, including the amount of noise reduction achieved, its response to genuine changes in the measurement, and the effect of the sampling interval.

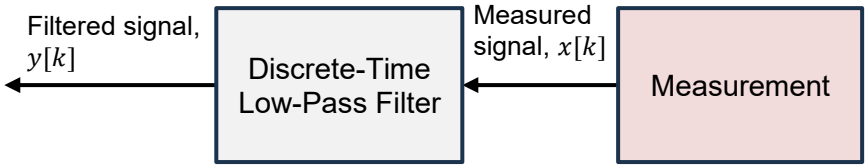


Figure 1: *Role of a Discrete-Time Low-Pass Filter in Measurement Processing*

The guide introduces the filter coefficients, explains how unity steady-state gain is maintained, investigates the effect of changing the filter coefficient, and considers practical issues including stability, sampling and filter initialisation. The overall aim is to provide a practical method for selecting a low-pass filter that improves the quality of a measured signal while preserving the information required for effective control.

3.0 Discrete-Time Low-Pass Filter Design

To design the filter, it is useful to begin with its discrete-time transfer function and then convert this into a difference equation that can be implemented directly. A first-order discrete-time low-pass filter can be written in the following form:

$$H(z) = \frac{a}{1 - bz^{-1}} \tag{1}$$

where a and b are the filter coefficients, and z^{-1} represents a delay of one sample. Equation (1) can be implemented in the following difference equation form:

$$y[k] = ax[k] + by[k-1] \tag{2}$$

where $x[k]$ is the current input to the filter and $y[k]$ is the current filtered output. For measurement filtering, $x[k]$ represents the current measurement value

supplied to the filter, for example the latest measurement obtained from a sensor. The term $y[k-1]$ represents the filtered output calculated at the previous sampling instant. Therefore, the filter calculates each new output using both the latest measurement and information retained from its previous filtered output.

Maintaining a steady-state gain of one

For measurement filtering, a constant measured value should produce the same value at the filtered output once the filter has reached steady state. For example, an input value of 1 should produce a steady-state filtered output value of 1. This is achieved by selecting the filter coefficients so that the steady-state gain is equal to unity. Considering Equation (1), this requires:

$$\frac{a}{1-b} = 1$$

Therefore:

$$a = 1 - b$$

Using the relationship in Equation (2), the difference equation can be written as:

$$y[k] = (1-b)x[k] + by[k-1] \tag{3}$$

This form ensures that a constant input is reproduced with the same amplitude in the steady-state filtered output. However, unity gain in this context refers specifically to a constant, or very slowly varying, input. It does not mean that all frequency components pass through the filter with a gain of one. The steady and slowly varying components of the measurement are largely retained, while faster variations are deliberately attenuated.

Equation (3) also provides a useful way of understanding how the filter operates. The term $(1-b)x[k]$ determines the contribution of the current measured input, while $by[k-1]$ determines the contribution of the previous filtered output. As these two coefficients add to one, changing b alters the balance between responding to the latest measurement and retaining information from the previous filtered output.

Selecting the Coefficient b

For practical low-pass measurement filtering, the coefficient b is typically selected such that:

$$0 < b < 1$$

A value of $b = 0$ is mathematically valid but produces no filtering because Equation (3) becomes $y[k] = x[k]$. It can therefore be used as a reference case when investigating the effect of the filter.

Increasing b provides greater smoothing of noisy measurement data but also causes the filtered signal to respond more slowly to changes in the measured

value. Conversely, decreasing b produces a faster response to changes in the measured signal, but provides less smoothing of measurement noise.

A larger value of b gives greater influence to the previous filtered output, whereas a smaller value gives greater influence to the current measured input.

Table 2 provides examples of filter designs for a range of coefficient values and illustrates the trade-off between measurement smoothing and response speed. Figure 2 presents the corresponding step responses for different filter coefficients. Each filter receives the same step input and uses the same sampling interval. A larger value of \(b\) produces a smaller initial change in the filtered output and a more gradual response over subsequent samples.

Table 2: *Effect of the Filter Coefficient b on the Discrete-Time Low-Pass Filter*

b	$a = 1 - b$	Current measured input / previous filtered output	Expected behaviour
0.00	1.00	100% / 0%	No filtering, as the filtered output equals the current input
0.50	0.50	50% / 50%	Light filtering with a relatively quick response
0.80	0.20	20% / 80%	Greater smoothing, with a slower response
0.90	0.10	10% / 90%	Strong smoothing, with greater reliance on the previous filtered output
0.95	0.05	5% / 95%	Very strong smoothing, but useful changes in the measurement may be noticeably delayed
0.98	0.02	2% / 98%	Very slow response; rapid but genuine changes in the measurement may be heavily attenuated

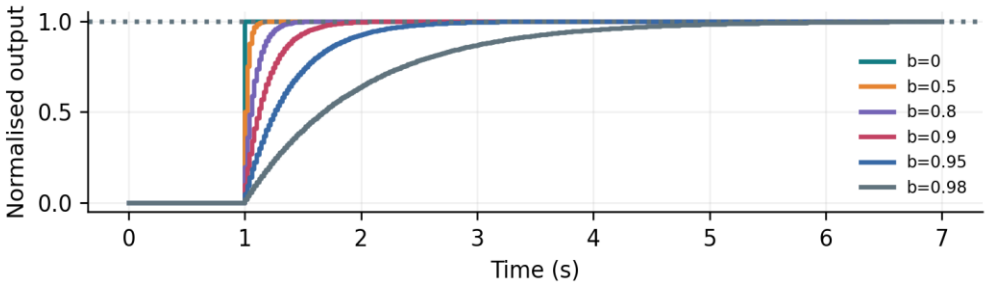


Figure 2: *Step Responses for Different Filter Coefficients*

For example, if $b = 0.8$, Equation (3) becomes:

$$y[k] = 0.2x[k] + 0.8y[k-1]$$

The new filtered output therefore contains a 20% contribution from the latest measured input and an 80% contribution from the previous filtered output. This causes sudden measurement changes to have a smaller immediate effect on the output, thereby providing smoothing.

If $b = 0.2$, the relationship instead becomes:

$$y[k] = 0.8x[k] + 0.2y[k-1]$$

The latest measured input now has much greater influence on the filtered output. The filter therefore responds more quickly to genuine changes in the measurement but provides less suppression of measurement noise.

Figure 3 presents noise smoothing and measurement delay for different coefficients. The same noisy measurement is processed using three coefficients. Increasing the coefficient suppresses more rapid variation while delaying the genuine step and the slower underlying changes. The uncorrupted reference signal is available in this example because the data have been constructed for demonstration purposes. In a practical experiment, the true signal will generally be unknown. Filter performance should therefore be assessed by considering both the reduction in unwanted variation during relatively steady operating conditions and the ability of the filter to retain genuine changes in the measured quantity.

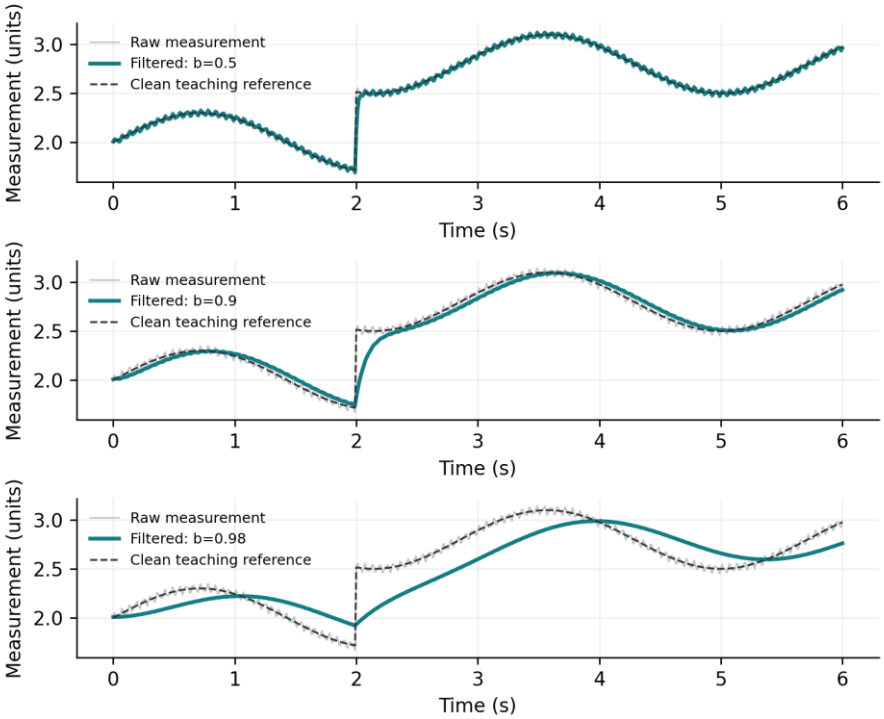


Figure 3: *Noise Smoothing and Measurement Delay for Different Filter Coefficients*

Filter Stability

For the filter to remain stable, its response should settle to a steady-state value following a change in the measured input, rather than continuing to increase or

oscillate without settling. For the first-order discrete-time low-pass filter considered here, this requires:

$$|b| < 1$$

For practical low-pass measurement filtering, b is normally selected such that:

$$0 < b < 1$$

Within this range, the influence of earlier filtered outputs gradually reduces over successive samples, allowing the filter output to settle towards the measured input.

A value of $b = 0$ is also mathematically stable, but Equation (3) reduces to:

$$y[k] = x[k]$$

and therefore provides no filtering. However, this case can be useful when comparing the unfiltered and filtered measurements.

Sampling Interval

The low-pass filter is evaluated once every sampling interval, T_s. Although T_s does not appear explicitly in the difference equation, it determines how frequently the measured input is sampled and the filtered output is updated.

The coefficient b determines the amount of smoothing from one sample to the next. However, the sampling interval T_s determines how much real time passes between these calculations. Consequently, the same value of b can produce different real-time filter behaviour when used with different sampling intervals. If the sampling interval is changed, the filter response should therefore be reassessed and b adjusted where necessary.

Initialisation

The low-pass filter requires the previous filtered output, $y[k-1]$, when calculating each new output. At start-up, however, no previous filtered output is available.

A simple practical approach is to initialise the filter using the first available measured input:

$$y[0] = x[0]$$

The first filtered output therefore begins at the measured value.

This is particularly useful when the measured quantity is already non-zero when the filter starts. Initialising the filter output to zero in this situation would cause the filtered output to move gradually from zero towards the measured value, creating an unnecessary start-up transient.

4.0 Summary

This learning guide has introduced the design and implementation of a first-order discrete-time low-pass filter for reducing high-frequency variation in measured signals. The filter can be implemented directly using a difference equation, making it suitable for discrete-time control systems and microcontroller-based applications.

By maintaining the relationship $a = 1 - b$, the filter has a steady-state gain of one so that constant and slowly varying measurements retain their expected amplitude. The coefficient b determines the balance between the current measured input and the previous filtered output. Increasing b generally provides greater measurement smoothing but also produces a slower response to genuine changes in the measured signal.

The filter coefficient must therefore be selected as a practical compromise between noise reduction and measurement responsiveness. The sampling interval must also be considered because it determines how frequently the filter is evaluated and therefore affects the filter behaviour in real time. If the sampling interval is changed, the filter response should be reassessed.

For practical implementation, the filter should also be correctly initialised. Setting the initial filtered output equal to the first measured value can prevent an unnecessary start-up transient when the measured quantity is already non-zero.

The suitability of a low-pass filter should ultimately be assessed using measured data from the application. The objective is not to produce the smoothest possible signal, but to reduce unwanted measurement variation while retaining the genuine changes required to understand and control the physical system.

Recommended Citation

If you have used this learning guide in your teaching, learning or research, please cite it using the reference below, adapting the format as required by your preferred referencing style.

> Pickering, J. E. (2026). Discrete-Time Low-Pass Filter Design. ACE-Lab. Teaching Notes: ACE-Lab Learning Guide, Version 1.0. https://zenodo.org/records/23217095